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Analysis of the ICT sector using the synthetic control method

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Abstrakt

Tato práce využívá metodu syntetické kontroly k vyhodnocení kauzálního dopadu referenda o brexitu na sektor vědy a technologií ve Spojeném Království. Pomocí pečlivě vybraného vybraných prediktorů a donorového souboru složeného ze zemí OECD je sestavena tzv. syntetická Velká Británie, která slouží jako kontrafaktuální referenční trajektorie pro vývoj sektoru bez referenda. Porovnáním skutečného vývoje a syntetické trajektorie, analýza odhaluje statisticky významný krátkodobý negativní dopad na hrubou přidanou hodnotu sektoru, přičemž rozdíl se stabilizuje po roce 2020. Přestože dlouhodobé důsledky zůstávají nejisté, zjištění ukazují, že referendum o brexitu mělo měřitelný ekonomický dopad – minimálně v krátkodobém horizontu – na britskou inovační ekonomiku.

Klíčová slova

Metoda Syntetické Kontroly, Brexit, sektor vědy a technologií, Spojené království

Abstract

This thesis applies the Synthetic Control Method to evaluate the causal impact of the Brexit referendum on the Science and Technology sector in the United Kingdom. Using a carefully selected set of predictors and a donor pool comprised of OECD countries, a synthetic UK is constructed to act as a counterfactual benchmark to the real UK. By comparing the two, the research shows a statistically significant negative impact on sector's Gross Value Added in the short term, with the gap stabilizing after 2020. While the long-term effect remains uncertain, the findings demonstrate that Brexit imposed a measurable economic cost—at a minimum in the short term—on the UK's innovation economy.

Keywords

Synthetic Control Method, Brexit, Science and Technology sector, United Kingdom

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Introduction

The UK has one of the most developed economies in Europe, built on innovation, advanced manufacturing, and services (Abdulla et al., 2025). It is considered a European leader in research and development (R&D), supported by a strong higher education system, a dynamic start-up ecosystem, and being the financial hub of Europe (Abdulla et al., 2025; UK, 2023). The UK has been deeply integrated within the European Union (EU) for more than 40 years, before deciding to leave in the 2016 Brexit referendum (the 2016 public vote on the United Kingdom's membership in the European Union).

While there has been a vast and educated debate about Brexit's effect on the whole UK economy, determining the impact on GDP, imports, households, and productivity of firms, much less has been researched about the impact on the Science and Technology sector of the United Kingdom. This sector represents approximately 14% of the total GDP of the UK and is the driving force of growth in recent years and is comprised of industries such as IT, scientific research and development, manufacturing of computer and other electronic components, telecommunications, or life sciences (OECD, 2025a; Taylor, 2024). This thesis aims to fill this gap by providing a more granular view on how political events, such as the Brexit referendum, can impact an R&D-intensive industry. The findings contribute to understanding the broader sector-specific consequences of Brexit and may serve as a valuable input for future policy discussions related to innovation, technology, and science, as well as offer insight into the potential impact of exiting the European Union for other members.

The main objective of this thesis is to estimate the causal impact of the 2016 Brexit referendum on the Science and Technology sector of the United Kingdom. The thesis uses the Gross Value Added (GVA) of the sector as the underlying outcome indicator. The aim is fulfilled by constructing the counterfactual UK considering the Brexit referendum never happened and then analyzing the difference between the real development and the counterfactual development. The counterfactual is constructed using the Synthetic Control Method.

Firstly, this thesis offers a review of research done estimating the impact of Brexit referendum on the UK economy and suggests where this thesis positions itself among the literature. It also illustrates the wide range of possible applications of the Synthetic Control Method, showing that it can be meaningfully used to assess the impact of major policy events. Subsequently, the methodological approach is described theoretically in detail to allow replicability of the study. Third, the results of the research are presented and discussed, along with various robustness and diagnostic checks to provide credibility to the taken approach. Lastly, possible contextual economic reasons are suggested, and conclusions are drawn.

1. Existing Literature

Brexit's effect on UK economy has been a vibrant topic in the last decade. Many studies have been made even before the Brexit referendum in 2016 trying to predict what impact Brexit would have had on UK economy, with often very different conclusions (Busch & Matthes, 2016). These studies typically looked at the UK's GDP outlook with or without Brexit using their own economic models such as CEBR (2015) or Dhingra et al. (2016), or were looking backward to the effects EU membership had on UK economy, for example Crafts (2016) or Cuaresma et al. (2008). The general consensus was, according to Busch and Matthes (2016), that Brexit would have caused a moderate and manageable net loss below 5% of the UK's GDP in the long term.

However, Busch and Matthes (2016) also suggest that results from the forward-looking projections are often underestimated and that maybe a better way to judge the effect is to look back and judge the effect of joining the EU. Looking at these studies suddenly, the results look more pessimistic as, for example, a backward looking study from Crafts (2016) showed a possible 10% decline in GDP if UK leaves the European Union. Studies from Campos et al. (2014) and Campos et al. (2019) use the synthetic control method to estimate whether there is a positive causal effect on the country's GDP after joining the EU, finding that there is indeed a strong positive effect in the low tens of %, suggesting a possible larger negative effect on the UK's GDP when leaving the EU.

All this literature worked only with data before the Brexit referendum and could only work with predictions and estimation of how the UK economy is going to react after the Brexit referendum and consequently the Brexit itself. The remaining literature is all post-referendum, assessing the impact of the Brexit referendum through different empirical approaches which is also the case of this thesis.

Born et al. (2019) determine the effect of the Brexit referendum using the Synthetic Control Method and using UK GDP as the output variable. They argue that the unexpected result of the Brexit referendum provides a unique opportunity to determine a causal effect. Also, they incorporate various robustness tests to justify their casual interpretation and find that the Brexit referendum caused a reduction of UK GDP by 1.7 - 2.5% by the end of 2018. They go further and zoom in to see what behavior after the referendum led to this decline and find out that firms and households have lowered their consumption and investments after the referendum in anticipation of the Brexit. The results of the study also suggest that a large part of the impact is attributable to uncertainty. The authors themselves propose that it would have been beneficial to redo or replicate their study in the future, as approximately a year and half of post-intervention data provides only a limited view on the impact, and looking at the impact of the Brexit referendum after a longer period could result in more detailed results.

A study by Bloom et al. (2019), endorsed by the Bank of England, looks at the impact the

Brexit referendum had from a microeconomic perspective, using data from a nation-wide survey. They firstly support the assumption that the result of the Brexit referendum was largely unexpected and therefore a causal effect can be measured. They use a difference-in-difference estimator to estimate the impact and find that the high uncertainty and anticipation of Brexit led to reduced investments by 11% after three years after the referendum and estimate that it also led to a decrease in the productivity of UK firms by 2%-5%. They argue that the majority of this drop was caused by a negative within-firm effect possible caused by firms committing several hours a week of their c-level executives' time to Brexit planning.

Some studies wanted to explore the uncertainty element after the Brexit referendum in more depth, Steinberg (2019) examines the macroeconomic impact of uncertain trade policy after the Brexit referendum, using a dynamic, stochastic, general equilibrium model. He uses the model to construct benchmark UK economy in which the Brexit never happens and also to model possible outcomes of the uncertainty "shocks" after the Brexit referendum. Steinberg finds that the total consumption welfare cost for UK households is between 0.4 and 1.2% and the uncertainty about trade policy accounts for less than 0.25% of the cost.

Not all studies about the impact of Brexit referendum look at output indicators like GDP, consumption or productivity, several studies look at other economical indicators. For example, Papyrakis et al. (2023) determines the impact on UK employment using the Synthetic Control Method or Breinlich et al. (2020) study the impact on foreign direct investment also using the Synthetic Control Method. Breinlich et al. (2020) find out that Brexit referendum increased the number of foreign direct investment transactions UK firms made in EU countries. This increase was isolated as no increase was detected in transactions made by UK firms outside of EU and also no increase was observed in EU firms investing in UK which suggests UK firms were expanding their operations outside of UK, indicating reduced confidence in UK's economic outlook.

Opatrny (2021) uses the Synthetic Control Method to estimate the impact of the Brexit referendum on UK financial markets, specifically the UK stock exchange index, long-term government bonds, and exchange rates. The study finds that there was no significant change in the index, but finds a strong negative effect of 1.2 percentage points on the 10-year bond yield. The results are inconclusive about the exchange rate as the synthetic control is not robust for this use-case.

Consistently throughout the studies, we see that Brexit had a negative impact on the UK's economy. Lowering the GDP, consumption, and investment of households and firms, lowering the productivity of the firms, lowering the 10-year bond yield, and increasing the FDIs of UK firms in the EU signifying loss of confidence in the UK economy, all probably because of uncertainty of what Brexit is going to bring and anticipation of negative UK's economical outlook. Many different approaches are used to estimate the effect, with the Synthetic Control Method being among the most prevalent.

To further showcase the use of the Synthetic Control Method these are some of the most influential papers using the method not concerned with Brexit; Abadie and Gardeazabal

(2003) where the method is proposed for the first time estimating the economic cost of a terrorist conflict; Abadie et al. (2010) estimating the effect of a tobacco control program in California; Abadie (2021) discussing the proper use of the method; Xu (2017) generalizing the use of the Synthetic Control Method to use on causal inference with interactive fixed effects models or Billmeier and Nannicini (2013) who use the method to assess the impact of economic liberalization policies on the real GDP of different countries. The wide range of possible applications of the Synthetic Control Method demonstrates its strength as an approach for causal inference in comparative case studies.

Looking at the literature about what effect the Brexit referendum had on the Science and Technology sector, one can only find anecdotal evidence on the impact such as Sellin and Mishra (2024) suggesting a negative relationship between uncertainty and innovation, or Wilson (2019), suggesting a negative effect on forensic science and technology, or Pardy and Ampudia (2023) who use the difference-in-difference method on survey data and find out that SMEs scale back on their adoption of digital technology after Brexit referendum. As to date no comprehensive study has evaluated the effect of the Brexit referendum on the UK's Science and Technology sector, this thesis aims to fill that gap.

2. Methodology

To reliably evaluate the effect of the Brexit referendum on the Science and Technology sector of the United Kingdom a Synthetic Control approach proposed by Abadie and Gardeazabal (2003) and Abadie et al. (2010) is selected as it provides a reliable and data-driven way to construct an appropriate counterfactual comparison economy. This counterfactual should track the selected indicator of the UK as precisely as possible prior to the referendum and should not be affected by the result of the Brexit referendum. In other words, the underlying assumption is that if there was no referendum or the referendum ended with the UK staying in the EU, the UK's and the comparison economy's indicator would have followed the same path even after 2016. By comparing the outcome indicator of these two economies after the Brexit referendum, we can quantify the effect the referendum had on the UK Science and Technology sector. Principles and requirements summed up by Abadie (2021) are followed.

2.1 Synthetic Control Method

2.2 The synthetic control estimator

Let $J + 1$ be the countries observed over T time periods. Without loss of generality, let country $j = 1$ denote the treated unit (UK), then countries $j = 2, \dots, J + 1$ form the donor pool which is the set of all possible comparison units not affected by the Brexit referendum. For each unit $j \in \{1, \dots, J + 1\}$ and time $t \in \{1, \dots, T\}$, an outcome variable Y_{jt} is observed.

The synthetic control is based on the observation that the behavior of the treated unit, can be approximated by a combination of the units from the donor pool. The synthetic control is the weighted average of the units from the donor pool:

$$\hat{Y}_{1t}^N = \sum_{j=2}^{J+1} w_j Y_{jt}, \quad (2.1)$$

where $\hat{Y}_{1t}^N$ is the outcome for the treated unit without the intervention, later called *doppelganger*, *counterfactual* or *synthetic control*, and w_j are the individual weights for j units from the donor pool. To avoid extrapolation, $w_j \geq 0$ and $\sum_{j=2}^{J+1} w_j = 1$. The gaps between the synthetic control and actual outcome of the treated unit, which quantify the effect of the intervention, are defined as:

$$\hat{\tau}_{1t} = Y_{1t} - \hat{Y}_{1t}^N. \quad (2.2)$$

Let T_0 , which $0 < T_0 < T$, be the number of pre-intervention periods. Then Equation (2.2), where $t \in \{T_0 + 1, \dots, T\}$ quantify the estimated effect of the intervention.

The vector of weights $\mathbf{W} = (w_2, \dots, w_{J+1})'$ ¹ in Equation (2.1), is chosen such that the synthetic control best matches the pre-intervention, $t \in \{1, \dots, T_0\}$, characteristics of the treated unit. These characteristics, commonly referred to as predictors of the outcome variable of the treated unit, should not themselves be affected by the treatment (Abadie, 2021). Assume a set of K predictors for each unit $X_{1j}, \dots, X_{Kj}$ in which the pre-intervention values of the outcome variable can also be included. The vector of optimal weights, $\mathbf{W}^* = (w_2^*, \dots, w_{J+1}^*)'$, is then obtained by minimizing the distance between $\mathbf{X}_1$, the treated unit's $(K \times 1)$ vector of pre-intervention characteristics, and $\mathbf{X}_0 \mathbf{W}$, the weighted combination of the donor pool characteristics where $\mathbf{X}_0$ is a $(K \times J)$ matrix of the same predictors as in $\mathbf{X}_1$ but for all units in the donor pool. Specifically, we minimize:

$$\|\mathbf{X}_1 - \mathbf{X}_0 \mathbf{W}\|_{\mathbf{V}} = \sqrt{(\mathbf{X}_1 - \mathbf{X}_0 \mathbf{W})' \mathbf{V} (\mathbf{X}_1 - \mathbf{X}_0 \mathbf{W})}, \quad (2.3)$$

where $\|\cdot\|_{\mathbf{V}}$ is a weighted norm weighted by the $\mathbf{V}$ matrix, which is a $(K \times K)$ positive semidefinite symmetric matrix reflecting the relative importance of each predictor in approximating the treated unit. Abadie and Gardeazabal (2003) and Abadie et al. (2010) propose a data-driven procedure to choose $\mathbf{V}$ so that $\mathbf{W}(\mathbf{V})$ minimizes the mean squared prediction error (MSPE) of the outcome variable in the pre-intervention period:

$$\sum_{t=1}^{T_0} \left(Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt} \right)^2. \quad (2.4)$$

Where $\mathbf{W}^* = \mathbf{W}(\mathbf{V}^*) = (w_2^*, \dots, w_{J+1}^*)$ and $\mathbf{V}^*$ is the optimal estimated matrix $\mathbf{V}$, reflecting predictors' relative importance. After the synthetic control is constructed, the pre-treatment MSPE can be computed using a common formula for MSPE:

$$\text{MSPE} = \frac{1}{T_0} \sum_{t=1}^{T_0} \left(Y_{1t} - \hat{Y}_{1t} \right)^2. \quad (2.5)$$

MSPE serves as a benchmark for evaluating the credibility of synthetic control. A low MSPE suggests that the synthetic control provides a good fit in regards to the real behavior of the outcome variable. MSPE is also used to test the estimated effect of the treatment in the post-intervention period by comparing the pre-intervention MSPE and the post-intervention MSPE with other units from the donor pool. After obtaining the optimal weights, $\mathbf{W}^*$, the estimated treatment effect for the treated unit is:

$$\hat{\tau}_{1t} = Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt}, \quad \text{for } t \in \{T_0 + 1, \dots, T\}. \quad (2.6)$$

¹Throughout this thesis, vectors are denoted using bold capital letters, consistent with the notation adopted by the authors of the Synthetic Control Method (Abadie, 2021; Abadie & Gardeazabal, 2003; Abadie et al., 2010), even though it differs from standard vector notation. This choice ensures consistency with the referenced methodology.

2.3 Inference

To test whether the effect of the intervention is statistically significant, Abadie et al. (2010) propose a test statistic based on permutations. The test calculates a p-value from the ratio of Root Mean Squared Prediction Errors (RMSPEs) after and before treatment.

Firstly, both RMSPEs are calculated for all countries, similar to the Equation (2.5), but square-rooted and generalized for two general time periods. For $0 \leq t_1 \leq t_2 \leq T$ and $j \in \{1, \dots, J+1\}$, let:

$$R_j(t_1, t_2) = \left(\frac{1}{t_2 - t_1 + 1} \sum_{t=t_1}^{t_2} (Y_{jt} - \hat{Y}_{jt}^N)^2 \right)^{1/2}, \quad (2.7)$$

where $\hat{Y}_{jt}^N$ denotes the outcome for country j at time t as predicted by its synthetic control, constructed by taking country j as the treated one and using all other countries as the donor pool. In our case to compare the pre-intervention and post-intervention RMSPEs, $t_1 = 1$ and $t_2 = T_0$ for the pre-intervention RMSPE, and $t_1 = T_0 + 1$ and $t_2 = T$ for the post-intervention RMSPE. The RMSPE ratio for each unit is calculated as follows:

$$r_j = \frac{R_j(T_0 + 1, T)}{R_j(1, T_0)}, \quad (2.8)$$

r_j is used to quantify the fit of the synthetic control for country j after the Brexit referendum relative to the pre-referendum period. The p-value is then calculated based on permutation distribution of r_j :

$$p = \frac{1}{J+1} \sum_{j=1}^{J+1} I_+(r_j - r_1), \quad (2.9)$$

where $I_+(\cdot)$ is an indicator function returning one for non-negative arguments and zero otherwise. This Equation is simply put, dividing the amount of synthetic controls of donor pool countries that have a higher r_j than the UK's synthetic control by the total number of countries. The p-value can then be commonly compared with the significance levels to test the statistical significance of the effect of the intervention.

2.4 Robustness of the method

Abadie (2021) criticizes a frequent misuse of the method in empirical research, highlighting that it often leads to misleading conclusions, and therefore proposes a set of diagnostic and robustness checks to judge the credibility of the method's application. Three tests are used in this thesis to judge the credibility of the constructed synthetic control and to test the sensitivity of the counterfactual to changes in the study design. In-time placebo test, placebo test across units and Leave-one-out test are carried out to evaluate it.

2.4.1 In-time placebo test

In this test, the treatment is artificially back-dated to years prior to the actual treatment. The artificial treatment year, $T'_0 + 1$ should be before the actual one, $T_0 + 1$ and thus this has to be fulfilled, $0 < T'_0 < T_0$. Two things are evaluated when using this test.

The first thing to look at is that the synthetic control, $\hat{Y}_{1t}^N$, should still closely track the real development of the treated unit before the actual treatment and after the artificial one, specifically in this period $\{T'_0 + 1, \dots, T_0 + 1\}$. According to Abadie et al. (2010) & Abadie (2021) if this is achieved, the synthetic control gains credibility because it shows that it is capable of reliably reproducing the trajectory of the treated unit, suggesting that any effect observed in the post-treatment period is caused by the treatment and not by a potential lack of predictive power of the synthetic control. In addition, it validates that there is no anticipation effect that would bias the synthetic control estimator (Abadie, 2021).

Secondly, a visible gap between the doppelganger and the real development should emerge around the time of the actual treatment for all the backdated synthetic controls (Abadie, 2021). If this is fulfilled, the test increases the credibility of the synthetic control and the observed effect (Abadie, 2021).

2.4.2 Cross unit placebo test

The second placebo test makes the synthetic control for each country from the donor pool in the exact same way as it was done for the treated unit. Formally, for each country $j \in \{2, \dots, J+1\}$, we estimate a synthetic outcome $\hat{Y}_{jt}^N$ using a synthetic control constructed from all other countries $i \neq j$ in the donor pool:

$$\hat{Y}_{jt}^N = \sum_{\substack{i=1 \\ i \neq j}}^{J+1} w_i^{(j)} Y_{it}. \quad (2.10)$$

The resulting gaps between actual and synthetic outcomes are then compared to the actually treated unit. If the gaps of the treated unit dominate the gaps of similarly well fit placebo synthetic control, it supports a causal interpretation of the treatment's effect (Abadie, 2021) & (Born et al., 2019).

2.4.3 Leave-one-out test

To test whether the synthetic control is robust to different selection of the donor pool, a leave-one-out test is performed. It tests how sensitive the results are when restricting the donor pool. Each country that contributed significantly to the construction of the synthetic control is left out of the donor pool at a time, and the synthetic control is reconstructed.

All the synthetic controls are then compared to the synthetic control from the unrestricted donor pool, and ideally no significant deviation is seen.

If leaving one of the units out of the donor pool alters the estimated effect significantly while the pre-intervention fit remains largely unchanged, it should be investigated further whether the change is due to some other interventions on the excluded unit or whether it is due to some unit-specific shocks. When interpreting, the significant effect of excluding one of the countries must be borne in mind.

2.5 Data

The application of the synthetic control method requires data about the predictors and the outcome variable for both the treated unit and the units from the donor pool. The donor pool was built only from the OECD countries to ensure a homogeneous group. Countries that had all missing values for some predictors were excluded. The remaining countries had a maximum proportion of missing values of 28.5%. The missing values were linearly interpolated in the pre-treatment period as the basic Synthetic Control Method requires a complete dataset with no missing values in the predictor and outcome variables during the pre-treatment period (Abadie, 2021). This data-processing must be considered when interpreting the results but does not apriori disregard the method since the predictors are either way averaged. To make the synthetic control completely robust against missing values, one can, for example, apply a model proposed by Amjad et al. (2018), but it is not used in this thesis.

The final donor pool consists of 31 countries, and since no additional restrictions are imposed on the donor pool, the construction of the synthetic control is therefore entirely data driven. Nevertheless, a robustness analysis is conducted afterwards to assess the sensitivity of the results to different approaches to constructing the donor pool.

All variables were observed on a yearly basis from 1995 to 2024 using data published by OECD (2025a). The intervention, the Brexit referendum, occurred in 2016, marking the beginning of the post-treatment period.

Gross Value Added per thousand people (GVA) was used as the outcome indicator as it provides a clear and quantifiable measure of the sector's contribution to the aggregate economic output (OECD, 2025a). GVA reflects the value of goods and services produced, net of intermediate consumption, and therefore provides a suitable measure of sectoral productivity and performance (OECD, 2025b). It also offers a consistent basis for cross-country comparison, which is essential to constructing the doppelganger from Equation (2.1) (Abadie, 2021).

Nine predictors were chosen from the available data on the sector in the OECD data set database. The chosen indicators must have been tied to the Science and Technology sector, include the observed years and were scaled to per capita or other unit that allowed the

indicators to be compared across countries. The following predictors were selected to ensure a close systematic match between the UK and its synthetic counterpart at other characteristics and not only the GVA so the two economies are comparable between each other.

- **G_XPOP** – Gross domestic expenditure on R&D (GERD) per capita in current PPP dollars
- **GV_XGDP** – GERD as a percentage of GDP
- **G_FBXGDP** – GERD financed by the business enterprise sector as a percentage of GDP
- **BP_TTXTT** – Business enterprise R&D personnel as a percentage of the national total
- **TD_XCOMP** – Export market share in the computer, electronic and optical industry
- **P_ICTPCT** – Number of patent applications to the PCT in ICT sector per thousand employees in R&D
- **GVA** – Gross Value Added by the sector per 1000 people
- **TD_XCOMP** (mean 2012–2015) – Average export market share in the computer, electronic and optical industry
- **B_XGDP** (mean 2012–2015) – Average GERD financed by government as a share of GDP

The mean of these predictors and the outcome variable are used to calculate the appropriate weights of different countries from the donor pool to construct the synthetic UK.

All data requirements proposed by Abadie (2021), namely sufficient pre-intervention information, sufficient post-intervention information, and available aggregate data on predictors and outcomes, are met, providing confidence in the use of the synthetic control method.

3. The effect of Brexit referendum on Science and Technology sector of UK

3.1 Constructing synthetic UK

The synthetic control is constructed from a donor pool of 31 OECD countries, variables are observed from 1995 to 2024 with the treatment occurring in 2016. The outcome variable is GVA of the Science and Technology sector per thousand people in the UK and nine predictors were chosen to ensure a fit between the doppelganger economy and the real economy in other characteristics as well.

Figure 3.1 shows how closely the synthetic UK follows the real UK in the pre-treatment period, where the gap between the two should be as small as possible. Also, it shows the doppelganger gap after treatment, which depicts the estimated effect the referendum had. One can see that the synthetic UK tracks the real UK quite closely during the pre-treatment period, the MSPE is **0.06067**, indicating a close match, with slight deviation around the year 2007. This deviation could be possibly attributable to the shock the GVA experienced due to the 2008 financial crisis and due to the different timings of these shocks across OECD countries. The fit is relativized and compared to synthetic controls of other countries, in section 3.3.2 to make a more educated judgment on the credibility of the synthetic control, for detailed description see Section 2.5.

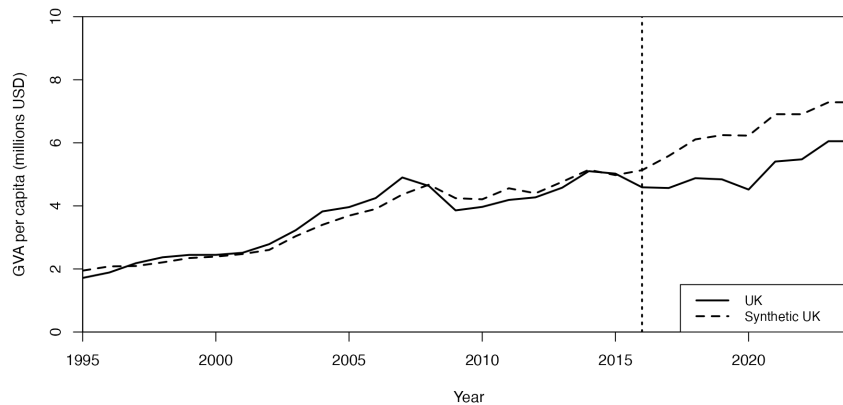


Figure 3.1: The development of GVA in the UK and its synthetic control. The vertical line marks the Brexit referendum in 2016.

The doppelganger gap is forming from the year 2016 and appears to be stabilizing around 2020. When interpreting the graph, it is important to keep in mind that the data are observed on yearly basis, which can distort the visual representation of timing. For instance,

the Brexit referendum occurred in Q2 of 2016 and could have influenced the second half of 2016. However, in Figure 2.1, it appears that the effect began even before the referendum. This is simply because all data points are measured at the end of each year and so is the referendum. As a result, when connecting the values from 2015 to 2016, we already observe a visible shift that may seem premature, but this is reasonable and likely reflects early reactions to the Brexit referendum in the second half of 2016.

The credibility of the synthetic control is supported by Table 3.1, which shows how well are the averages of the predictors matched between the doppelganger and UK. It shows that the synthetic control matches the UK economy also for other characteristics and not only the GVA. For example, Table 3.1 shows that the Gross Domestic Expenditure on R&D (GERD) as a percentage of GDP is approximately 17% for the synthetic control and around 20% for the real UK, ensuring similar dynamics for the two compared economies. This increases the confidence that the synthetic UK is well constructed.

Predictors	Units of measure	UK	Synthetic_UK
G_XPOP	GERD per capita (USD, PPP)	537.47	636.31
GV_XGDP	GERD as % of GDP	0.17	0.20
G_FBXGDP	Business-funded GERD (% of GDP)	0.76	0.91
TP_TTXEM	R&D personnel per 1000 employed	11.43	11.04
BP_TTXTT	Business R&D staff share (%)	48.21	58.11
P_ICTPCT	Patents in the sector per 1000 employees in R&D	3.57	4.36
GVA_per_capita	GVA per capita (million USD)	3.53	3.50
TD_XCOMP	Export share: computer/optical (%)	1.46	2.98
B_XGDP	BERD as % of GDP	1.45	1.21

Table 3.1: *Comparison of UK's and doppelganger's predictors*

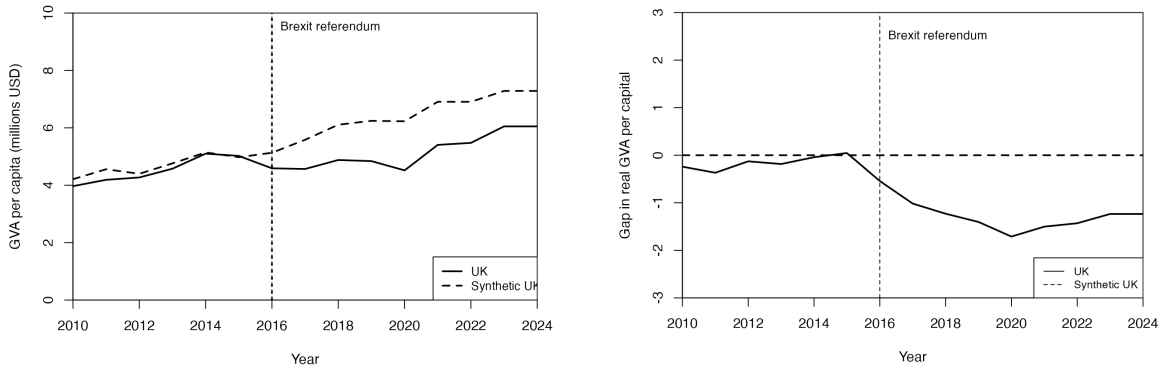
Table 3.2 presents the composition of the synthetic UK by showing the weights assigned to each country from the donor pool. The main component countries are Italy, United States, Ireland, and the Netherlands, respectively having weights of 0.40, 0.29, 0.23, and 0.08 (rounded to the second decimal point); Belgium and Greece are also included in the composition but have marginal weights which are lower than 0.01. The robustness of the synthetic control to the country selection is tested in section 3.3.3.

Country	Weight	Country	Weight	Country	Weight
Austria	<0.01	Finland	<0.01	Mexico	<0.01
Belgium	<0.01	France	<0.01	Netherlands	0.08
Canada	<0.01	Germany	<0.01	Norway	<0.01
Czechia	<0.01	Greece	<0.01	Poland	<0.01
Denmark	<0.01	Hungary	<0.01	Portugal	<0.01
Estonia	<0.01	Iceland	<0.01	Slovak Republic	<0.01
Ireland	0.23	Israel	<0.01	Slovenia	<0.01
Italy	0.40	Japan	<0.01	Spain	<0.01
Korea	<0.01	Latvia	<0.01	Sweden	<0.01
Lithuania	<0.01	Luxembourg	<0.01	Türkiye	<0.01
United States	0.29				

Table 3.2: *Weights assigned to donor pool countries*

3.2 The effect of Brexit referendum

To quantify the effect of the Brexit vote on the Gross Value Added outcome of the Science and Technology sector, one can simply compare the synthetic UK with the real development and look at gaps formed after the referendum. The significance and causality of the effect is tested and discussed in the following sections.



(a) *Zoomed visualization of the effect of Brexit referendum on the GVA* **(b)** *Gap between UK's and Synthetic UK's GVA*

Figure 3.2: *Visualization of the effect of Brexit referendum on GVA in the Science and Technology sector per 1000 people. Synthetic Control Results.*

There appears to be a substantial effect on the GVA after the Brexit referendum, Figures 3.2a, 3.2b. The difference, as can be seen in Figure (2.2), rises right after the referendum until the year 2020 where it seems the real UK returns to the trend before the referendum, and the difference stabilizes. The biggest gap between the doppelganger and the real UK

GVA output per 1,000 people is \$1.71 million in 2020 and stabilizes around the value \$1.24 million in the years 2023 and 2024. To put this into perspective, the difference stabilizes around \$84.4 billions of Gross Value Added by this sector, which means that the GVA of the real UK is 19% lower than the GVA reached by the doppelganger in 2024.

3.3 Robustness and causality of the method

To arrive to the desired result and declare with confidence that Brexit referendum did indeed have a negative effect on the GVA of the UK, this section provides placebo and robustness checks. According to Abadie (2021), the synthetic control estimator is credible and can be used to identify the causal effect as long as similar magnitudes are not estimated in cases where the intervention did not take place, both in time and across countries, and the synthetic control is robust against the study design.

3.3.1 In-time placebo test

The treatment is artificially backdated to years prior to the actual treatment to years 2009-2015, simulating seven artificial synthetic controls with the treatment set before the 2016 referendum. Figure 3.3 displays the results of this test. The shape and direction of the divergence between the synthetic control and real UK in Figure 3.2a closely matches those in Figure 3.3, even-though the magnitudes of the effect are somewhat bigger. This could vary due to a shorter pre-intervention period.

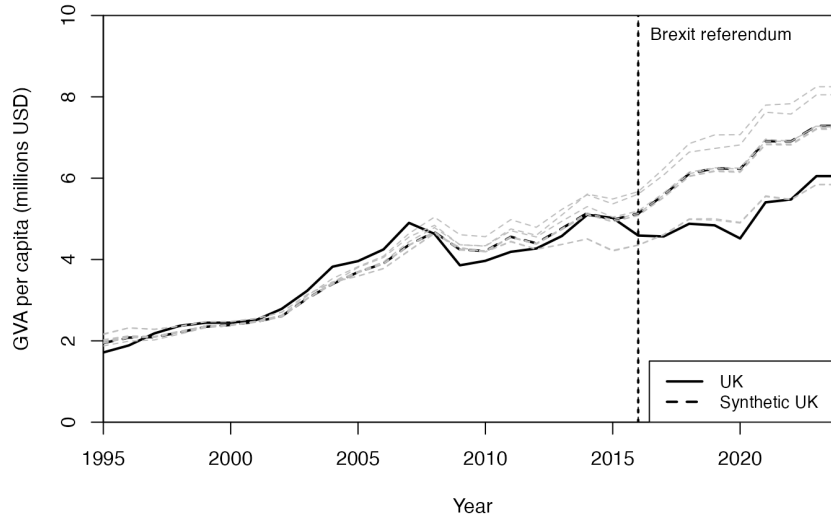


Figure 3.3: *In-time placebo test. It depicts the development of the GVA per 1000 people for real UK, black line; for the used synthetic control, dashed black line; and for placebo synthetic controls with backdated artificial treatments in gray dashed lines for years between 2009 and 2015.*

A noticeable deviation between the UK and its synthetic counterparts, apart for the 2012 placebo, consistently arises around 2016, even under placebo interventions occurring as early as 2009, Figure 3.3. This adds credibility to the interpretation that the observed post-2016 divergence reflects a genuine impact of the Brexit vote as the synthetic controls with backdated intervention both closely track the real development after the artificial treatment and before the actual one and a similar effect appears around similar time in the post-intervention period.

The synthetic control with a placebo treatment in 2012 has a different behavior than the rest, probably due to the selection of the special predictors that are the averages between the years of 2012 and 2015, and if we take only the value in 2012 for these predictors and we see a decline in the GVA in that year, we could possibly overfit the synthetic control, which then becomes non-informative. This was tested by running the same placebo test only with the special predictors taking mean firstly from 2013 till 2015 and then 2014 till 2015 and in these cases the "problematic" placebo years were respectively 2013 and then 2014, supporting this possibility.

The results of the in-time placebo test suggest that there is a causal negative effect of the Brexit referendum on the GVA of the UK. The synthetic placebo controls show a similar behavior of GVA after placebo treatments without the method specifying when the actual treatment occurred.

3.3.2 Cross country placebo test

Figures 3.4 and 3.5 visualize the cross-unit placebo test. They show synthetic control gaps for different countries which are standardized by the standard deviation of the gaps pre-treatment and are centered at year 2015 for better comparison. The highlighted black line are the gaps of the "original" synthetic control of UK.

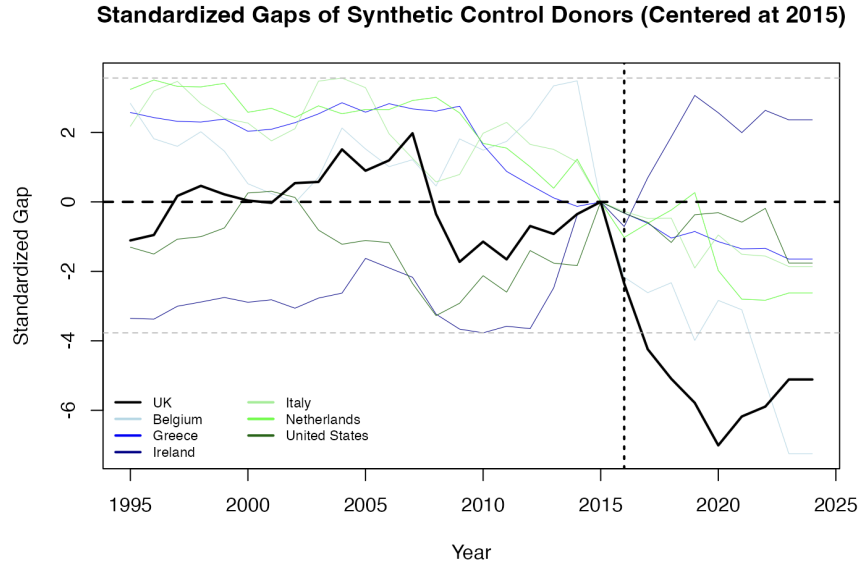


Figure 3.4: *Placebo test with synthetic control of used countries*

Figure 3.4 shows the comparison between the UK's gaps and the gaps of doppelgangers of the countries used to build the synthetic UK. First we see that the gaps are the lowest for the UK before the treatment which single-handedly does not say much about the credibility. Second, and that is more interesting, the UK's gaps dominate the gaps of the other countries until the year 2022 and the gaps become the most significant right after the vote in 2016, which adds to credibility and possible causality interpretation of the synthetic control. All donor countries until 2022 remain within the same limits, gray dashed horizontal lines in Figure 3.4, as before treatment. Around the years 2021 and 2022 there seems to be some strong negative effect on Belgium's GVA, this could have been caused for example, by the Covid-19 pandemic or some other shock, but this does not disregard our UK's synthetic control as the effect comes only after 5 years and is an isolated case among the donor countries.

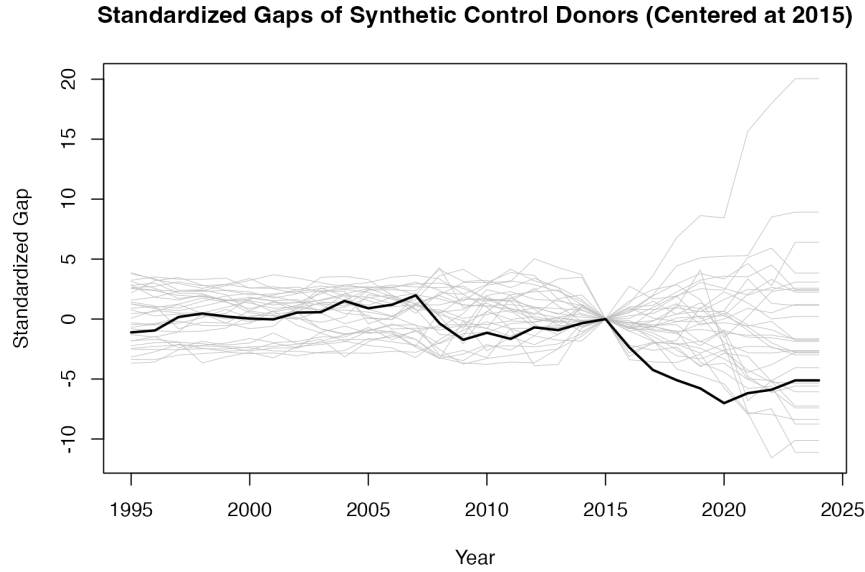


Figure 3.5: *Placebo test with synthetic controls of all countries from the donor pool. Black line depicts the standardized gaps between UK and UK's synthetic control. Grey lines show the standardized gaps between all the countries in the donor pool and their respective synthetic controls.*

Looking at Figure 3.5 which visualizes the gaps for all countries from the donor pool, one can see a similar behavior as in Figure 3.4, UK's gaps dominate gaps of the rest of the countries and then there are a few cases that dominate the UK's gaps after the year 2021. One gaps that stand out and reach the value of 20 are the gaps of Lithuania, which increased significantly after 2016, signifying a strong increase of the real GVA after the year 2016 which was not picked up by its synthetic control before 2016 and possibly suggesting that some big changes occurred around 2016 which positively impacted the Lithuanian Science and Technology sector. Lithuania experienced a startup boom between 2014 and 2024 when the startup economy increased 39-times, that is probably what we see in Figure 3.5 (Government, 2025). Since it is an isolated case and probably had nothing to do with Brexit it doesn't decrease the credibility of the synthetic control.

The results of the cross-unit placebo test indicate a strong negative short-term effect of the Brexit referendum on the UK's Science and Technology sector. The UK's doppelganger gaps dominate other gaps until the year 2021.

3.3.3 Leave-one-out

Another issue or bias that can arise is due to the selection of the donor pool countries. Figure 3.6 visualizes the results of the leave-one-out test. Comparing the actual synthetic control constructed from the full donor pool and the synthetic controls constructed from the constrained donor pools; shows that it is robust to leaving out all but one country.

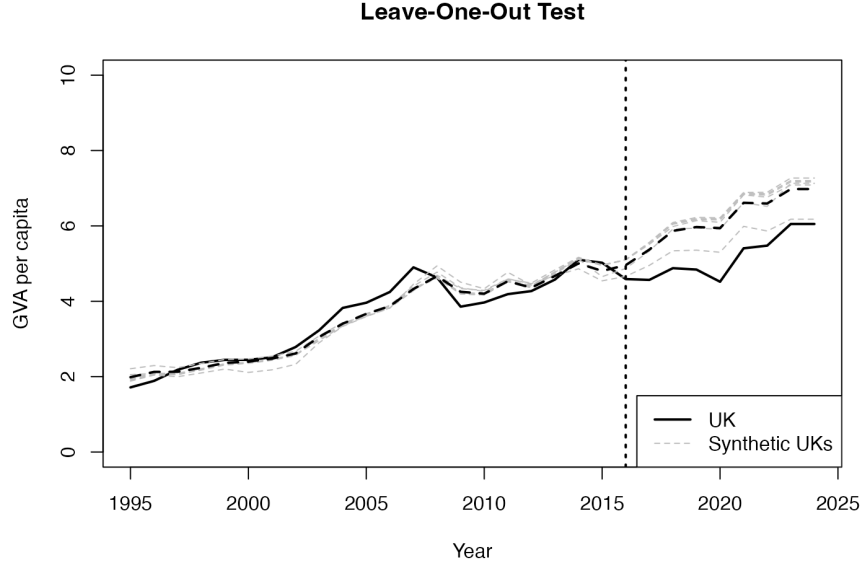


Figure 3.6: *Leave-one-out test for all countries that contributed to the construction of the synthetic control.*

The synthetic control used is sensitive to including Ireland in the donor pool. Ireland contributes with 22.80% to the construction of the doppelganger, Table 3.2. When Ireland is left out of the donor pool, the gap and, therefore, the effect of Brexit referendum appears to be smaller. This suggests that Ireland is driving the gap between the doppelganger and real UK, which is also supported by Figure 3.7 which compares the real GVAs of all the contributing countries with UK's. Since Ireland's GVA's trajectory is very similar to UK's between 2007 and 2012, the weight of Ireland is not the highest, it is not an economic outlier and the overall effect still persists even after leaving Ireland out of the donor pool, the inclusion of Ireland in the construction is reasonable and the results do not have to be disregarded but they have to be interpreted and justified with caution.

When taking into account the economic context, particularly the fact that many technological corporations had relocated to Ireland due to the Brexit referendum, the inclusion of Ireland in the donor pool becomes more justifiable (Jones, 2019). It can be argued that a significant portion of Ireland's GVA growth after 2016 was driven by these relocations, which is also supported by Figure 3.4 where the positive gaps between Ireland's actual GVA and its synthetic counterpart suggest that the Brexit referendum had a positive impact on Ireland's GVA. It is probable that if the outcome of the Brexit referendum was opposite, the technological corporations would have remained in the UK, thereby contributing to domestic growth instead. It is therefore justified to keep Ireland in the donor pool and let the synthetic control method use it for the construction of the doppelganger, which is also in line with the results from Born et al. (2019). However, it must be kept in mind that the results might be slightly overstated.

Overall, the leave-one-out test suggests that the synthetic control is robust to the study

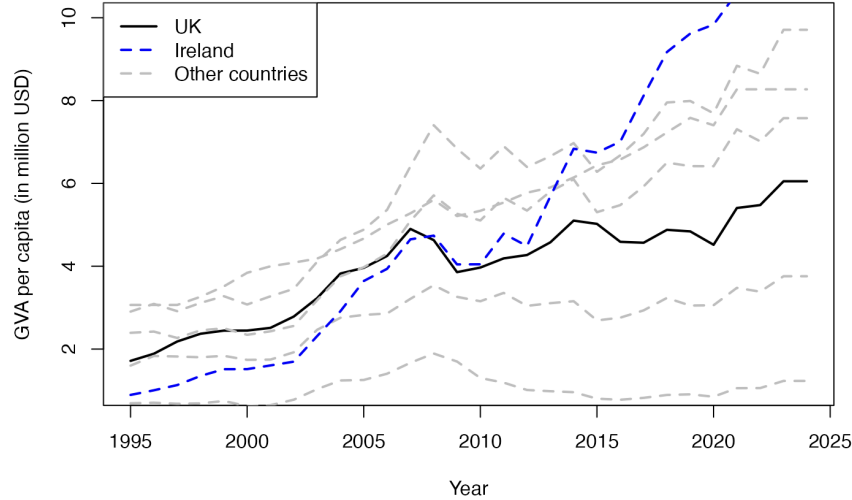


Figure 3.7: Comparison of the real development of the GVA in the countries that contributed to the construction of the synthetic control

design, but sensitive to the inclusion of Ireland in the donor pool; however, the sensitivity is justifiable with the economic context.

3.4 Inference

Building on top of the results from section 3.3.2, where placebo synthetic controls are constructed for all countries in the donor pool, statistical significance is assessed following the approach proposed by Abadie (2021).

The p-value of our synthetic control calculated for the entire post-intervention period is 0.161, indicating that the synthetic control is not statistically significant for the common significance levels of 0.05 or 0.10. But when the post-treatment period was bounded by 2021, as suggested from the discussion about Figure 3.5, the p-value jumps down to 0.032 making the estimated gaps statistically significant at the significance level of 0.05. The result is in line with the results from the cross-unit placebo test in section 3.3.2.

This result allows us to make a more educated conclusions about the results, suggesting that although the effect of Brexit referendum is diminishing over a longer period of time, the immediate effect after the referendum is statistically significant.

3.5 Results discussion

To sum up, after the robustness and placebo test, the synthetic control showed us that the Brexit referendum did indeed have a negative short-term effect on the UK's GVA with

inconclusive results on the long-term effect. Let's now look at the possible reason why this was the case.

Relocation of companies. According to Jones (2019), hundreds of companies, and mainly those in Science and Technology sector have relocated outside of the UK after the Brexit referendum. The most common destinations for relocating were Ireland, Luxembourg, Germany and Netherlands. Morris (2016) suggests it was mostly due to uncertainty the vote brought to the market and fear about attracting and retaining talent from the EU, trade barriers, or complicated schemes about data processing between the UK and the EU.

Decrease in investments. According to the data collected by the Bank of England business investments have been stagnating after the Brexit referendum diverging from the previous upward trend (Partington & correspondent, 2023). This is theorized to be again due to the uncertainty that spread after the Brexit referendum (Partington & correspondent, 2023). This is also supported by the results of studies by Born et al. (2019) and Breinlich et al. (2020).

Anticipation effect. Born et al. (2019) found out that after the Brexit referendum there was a significant decrease in private consumption growth, suggesting that households and firms had lowered their expenditures as they had expected, anticipated, the Brexit referendum to lower the UK's prosperity. Probably also lowering their spend on digital products which was seen among the SME's and their back scaling on digital adoption and tools (Pardy & Ampudia, 2023).

Depreciation of GBP. After the Brexit referendum, the British pound lost nearly 10% of its value in a single trading day and totally declined by around 15% against the euro and the US dollar. This depreciation led to a decrease in imports into the UK. This probably also affected firms in the Science and Technology sector for whom foreign imported services would be more expensive, thus having an impact on their overall output. The depreciation was, according to Plakandaras et al. (2017), also driven by uncertainty on the market.

Identifying the exact causes of the negative effect Brexit referendum had on the UK's GVA is almost impossible. The above are just some of the practical examples that occurred due to the referendum and somewhat probably contributed to the negative impact. Although a comprehensive analysis of the broader economic consequences of the Brexit referendum is left for future work, the findings in this thesis clearly demonstrate that the Brexit referendum had a significant short-term negative effect on the Science and Technology sector in the UK.

Conclusion

In conclusion, this thesis aims to estimate the causal impact of the 2016 Brexit referendum on the Science and Technology sector of the United Kingdom, which is fulfilled.

The counterfactual UK is reliably constructed from a donor pool of 31 OECD countries using a data-driven Synthetic Control Method. The credibility of the doppelganger is confirmed using the in-time placebo test, cross country placebo test, and the leave-one-out test. The only limitation revealed by the tests is that the synthetic control is sensitive to including Ireland in the donor pool, as it appears to significantly drive the estimated gap. However, if the economic context is taken into account, leaving Ireland in the donor pool can be justified.

It is found that the Brexit referendum had a significant immediate negative effect on the Gross Value Added by the Science and Technology sector. The estimated loss the sector has suffered due to the Brexit referendum is approximately \$84.4 billion, which is equivalent to losses of approximately 19%. The statistical significance of these results is tested using a permutation-based p-value. The results are statistically significant only if we look at the post-intervention period until 2021 ($p\text{-value} = 0.032$). When considering the entire post-intervention period, 2016 - 2024, the significance diminishes ($p\text{-value} = 0.161$). Thus, while the long-term consequences remain uncertain, the results provide compelling evidence of a substantial immediate economic cost. Several possible economic actions probably contributed to the measured impact: the relocation of firms to other EU countries, increased economic uncertainty, depreciation of the GBP, and reduction in investment in the UK by both households and firms due to the anticipation effect.

Overall, this thesis contributes to the broader literature on the economic consequences of Brexit by focusing on a Science and Technology sector that is often mentioned but rarely studied in depth. It helps to understand how radical political interventions can influence the prosperity of a specific sector.

References

- Abadie, A. (June 2021). Using Synthetic Controls: Feasibility, Data Requirements, and Methodological Aspects. *Journal of Economic Literature*, 59(2), 391–425. <https://doi.org/10.1257/jel.20191450>
- Abadie, A., Diamond, A., & Hainmueller, J. (June 2010). Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California’s Tobacco Control Program. *Journal of the American Statistical Association*, 105(490), 493–505. <https://doi.org/10.1198/jasa.2009.ap08746>
- Abadie, A., & Gardeazabal, J. (March 2003). The Economic Costs of Conflict: A Case Study of the Basque Country. *American Economic Review*, 93(1), 113–132. <https://doi.org/10.1257/000282803321455188>
- Abdulla, S., Schomberg, W., & Chopra, T. (May 2025). UK services sector contracts at steepest pace since 2023, PMI shows. *Reuters*. Retrieved May 6, 2025, from <https://www.reuters.com/world/uk/uk-services-sector-contracts-steepest-pace-since-2023-pmi-shows-2025-05-06/>
- Amjad, M., Shah, D., & Shen, D. (2018). Robust Synthetic Control. *Journal of Machine Learning Research*, 19(22), 1–51. Retrieved April 20, 2025, from <http://jmlr.org/papers/v19/17-777.html>
- Billmeier, A., & Nannicini, T. (2013). Assessing economic liberalization episodes: A synthetic control approach. *Review of Economics and Statistics*, 95(3), 983–1001.
- Bloom, N., Bunn, P., Chen, S., Mizen, P., Smietanka, P., & Thwaites, G. (September 2019). The Impact of Brexit on UK Firms. <https://doi.org/10.3386/w26218>
- Born, B., Müller, G. J., Schularick, M., & Sedláček, P. (May 2019). The Costs of Economic Nationalism: Evidence from the Brexit Experiment* [eprint: <https://academic.oup.com/ej/article-pdf/129/623/2722/31169510/uez020.pdf>]. *The Economic Journal*, 129(623), 2722–2744. <https://doi.org/10.1093/ej/uez020>
- Breinlich, H., Leromain, E., Novy, D., & Sampson, T. (May 2020). Voting with their money: Brexit and outward investment by UK firms. *European Economic Review*, 124, 103400. <https://doi.org/10.1016/j.euroecorev.2020.103400>
- Busch, B., & Matthes, J. (2016). Brexit - the economic impact: A meta-analysis [Number: 10/2016 Publisher: Institut der deutschen Wirtschaft (IW) / German Economic Institute]. *IW-Reports*. Retrieved May 3, 2025, from <https://ideas.repec.org/p/zbw/iwkrep/102016.html>
- Campos, N. F., Coricelli, F., & Moretti, L. (2014). Economic Growth and Political Integration: Estimating the Benefits from Membership in the European Union Using the Synthetic Counterfactuals Method. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.2432446>
- Campos, N. F., Coricelli, F., & Moretti, L. (May 2019). Institutional integration and economic growth in Europe. *Journal of Monetary Economics*, 103, 88–104. <https://doi.org/10.1016/j.jmoneco.2018.08.001>

- CEBR. (2015). Britain Stronger in Europe. Retrieved May 3, 2025, from <https://cebr.com/reports/britain-stronger-in-europe/>
- Crafts, N. (September 2016). Is Secular Stagnation the Future for Europe? In P. Askenazy, L. Bellmann, A. Bryson, & E. Moreno Galbis (Eds.), *Productivity Puzzles Across Europe* (p. 0). Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780198786160.003.0003>
- Cuaresma, J. C., Ritzberger-Grünwald, D., & and, M. A. S. (2008). Growth, convergence and EU membership [Publisher: Routledge _eprint: <https://doi.org/10.1080/00036840600749524>]. *Applied Economics*, 40(5), 643–656. <https://doi.org/10.1080/00036840600749524>
- Dhingra, S., Ottaviano, G., Sampson, T., & Reenen, J. V. (March 2016). The consequences of Brexit for UK trade and living standards [Number: 02 Publisher: Centre for Economic Performance, LSE]. *CEP Brexit Analysis Papers*. Retrieved May 3, 2025, from <https://ideas.repec.org/p/cep/cepbxt/02.html>
- Goverment, L. (January 2025). Lithuanian Startup Boom: Ecosystem Now Valued at €16 Billion. Retrieved April 19, 2025, from <https://lithuania.lt/governance-in-lithuania/lithuanian-startup-boom-ecosystem-now-valued-at-e16-billion/>
- Jones, H. (2019). Brexit fallout on UK finance intensifies - think tank | Reuters. Retrieved April 20, 2025, from <https://www.reuters.com/article/world/uk/brexit-fallout-on-uk-finance-intensifies-think-tank-idUSKBN1QS00A/>
- Morris, D. Z. (2016). Why Brexit is Bad for Tech Companies in the UK (And Everywhere Else). Retrieved April 19, 2025, from <https://fortune.com/2016/06/25/why-brexit-is-bad-for-tech/>
- OECD. (2025a). OECD Data Explorer • Annual GDP and components - output approach. Retrieved April 20, 2025, from https://data-explorer.oecd.org/vis?pg=0&bp=true&snb=11&tm=GVA&df%5Bds%5D=dsDisseminateFinalDMZ&df%5Bid%5D=DSD_NAMAIN10%40DF_TABLE1_OUTPUT&df%5Bag%5D=OECD.SDD.NAD&df%5Bvs%5D=2.0&dq=A.AUS.....V...&lom=LASTNPERIODS&lo=5&to%5BTIME_PERIOD%5D=false&lb=nm
- OECD. (2025b). Value added by activity. Retrieved April 20, 2025, from <https://www.oecd.org/en/data/indicators/value-added-by-activity.html>
- Opatrny, M. (May 2021). The impact of the Brexit vote on UK financial markets: a synthetic control method approach. *Empirica*, 48(2), 559–587. <https://doi.org/10.1007/s10663-020-09481-7>
- Papyrakis, E., Pellegrini, L., & and, L. T. (2023). Impacts of the brexit referendum on uk employment: A synthetic control method approach. *Applied Economics Letters*, 30(11), 1407–1410. <https://doi.org/10.1080/13504851.2022.2056567>
- Pardy, M., & Ampudia, D. (2023). *Brexit and Digital Technology Adoption of UK SMEs* (tech. rep.). Enterprise Research Centre. Retrieved May 3, 2025, from <https://www.enterpriseresearch.ac.uk/publications/brexit-and-digital-technology-adoption-of-uk-smes/>
- Partington, R., & correspondent, R. P. E. (November 2023). Brexit has ‘chilled’ business investment, says Bank of England deputy governor. *The Guardian*. Retrieved April 19,

- 2025, from <https://www.theguardian.com/business/2023/nov/21/brexit-has-chilled-business-investment-says-bank-of-england-deputy-governor>
- Plakandaras, V., Gupta, R., & Wohar, M. E. (2017). The depreciation of the pound post-brexit: Could it have been predicted? *Finance Research Letters*, 21, 206–213. <https://doi.org/https://doi.org/10.1016/j.frl.2016.12.003>
- Sellin, J., & Mishra, V. (2024). Assessing the impact of brexit on uk-based inventor teams' connectivity. *Journal of International Management*, 30(6), 101202. <https://doi.org/https://doi.org/10.1016/j.intman.2024.101202>
- Steinberg, J. B. (March 2019). Brexit and the macroeconomic impact of trade policy uncertainty. *Journal of International Economics*, 117, 175–195. <https://doi.org/10.1016/j.jinteco.2019.01.009>
- Taylor, R. (October 2024). Science and technology's contribution to the UK economy. Retrieved May 11, 2025, from <https://lordslibrary.parliament.uk/science-and-technologys-contribution-to-the-uk-economy/>
- UK, B. a. T. D. (2023). Advanced manufacturing plan (HTML version). Retrieved May 6, 2025, from <https://www.gov.uk/government/publications/advanced-manufacturing-plan/advanced-manufacturing-plan-html-version>
- Wilson, T. J. (2019). The impact of brexit on the future of uk forensic science and technology. *Forensic Science International*, 302, 109870. <https://doi.org/https://doi.org/10.1016/j.forsciint.2019.06.028>
- Xu, Y. (2017). Generalized synthetic control method: Causal inference with interactive fixed effects models. *Political Analysis*, 25(1), 57–76.